

SHIVAMSHU ROY

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EDUCATION

State University of New York at Buffalo

Master of Science in Engineering Science (Data Science)

Buffalo, NY

Aug 2026 – Expected May 2028

Bennett University

Bachelor of Technology in Computer Science (Data Science) CGPA: 8.93 / 10.0

Greater Noida, India

2020 – 2024

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C++, JavaScript, HTML/CSS

ML & Deep Learning: PyTorch, TensorFlow, Scikit-Learn, Time-Series (Holt-Winters, ARIMA), XGBoost, NLP, LLMs

Data & Cloud: Google BigQuery, PostgreSQL, MySQL, Firebase, FastAPI, Flask, Docker, ETL Pipelines

Analytics & Tools: Tableau, Power BI, Statistical Modeling, EDA, Git/GitHub, Linux/Bash, Jupyter

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems

PROFESSIONAL EXPERIENCE

Data Scientist

Dec 2024 – Feb 2026

Nxtwave Disruptive Technologies

Hyderabad, India

- Architected end-to-end data pipelines using SQL, BigQuery, and Python, automating cross-functional reporting and reducing manual data processing latency by 40%.
- Designed enterprise analytics dashboards querying 500K+ transactional student records; adopted across 4 departments, reducing cross-team reporting cycles by 35%.
- Built automated data integrity and anomaly detection workflows using Google Apps Script and REST APIs, cutting report turnaround time from 2 days to near real-time.

Data Science Intern

May 2024 – Aug 2024

Tata Consultancy Services (TCS)

Hyderabad, India

- Developed Python-based automated Exploratory Data Analysis (EDA) pipelines and statistical verification scripts, decreasing data cleaning and preparation time by 30%.
- Implemented scalable data transformation workflows and BI visualization reports to support operational KPI tracking across multi-disciplinary project teams.

PUBLICATIONS & RESEARCH

Stock Market Forecasting Using Advanced Time Series Analysis

[\[IEEE Xplore Link\]](#)

Published & Presented at the IEEE International Conference on Computer Vision and Machine Intelligence (CVMI-2023)

- Formulated optimized Holt-Winters exponential smoothing and adaptive moving average models for financial market index forecasting, achieving enhanced trend projection accuracy over baseline ARIMA architectures.
- Engineered specialized optimization algorithms to reduce forecast error metrics (RMSE/MAE) on volatile historical financial series.

PROJECTS

CheckmateLab – Browser-Based Chess Engine & Analytics

[\[GitHub\]](#) | [\[Live Demo\]](#)

- Engineered a client-side chess evaluation dashboard integrating Stockfish 16 NNUE via WebAssembly, delivering real-time depth evaluations, dynamic win probabilities, and opening book detection.
- Implemented move accuracy scoring algorithms and Firebase OAuth authentication to track real-time player ELO ratings.

ClearHire AI – Intelligent Application & Ghosting Risk Tracker

[\[GitHub\]](#)

- Developed an AI job application intelligence engine featuring Gmail OAuth synchronization to automatically parse status updates and correspondence from recruiters.
- Formulated a predictive "Ghosting Risk" scoring algorithm calculating company response velocity, status change indicators, and automated context-aware follow-up triggers.

Green Grid AI – Geospatial EV Charging Demand Prediction

[\[GitHub\]](#)

- Built machine learning regression models (Random Forest, Gradient Boosting) achieving an R^2 score of 0.88 to forecast municipal EV charging infrastructure demand across 17,000+ spatial data records.